THE ALGEBRA C = F(G_*O_U) AND THE RING MAP A -> C

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4574-4586 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4588-4643, the complete author proof (54 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_stacks-stacks-more-morphisms.json (pb=2250, stmt 4574-4586, proof 4588-4643).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-affine' also exists in stacks-descent, stacks-limits, stacks-spaces-limits, stacks-varieties — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GNS: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4643–4643. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GQY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4643–4643. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 077A: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4643–4643. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GQM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4643–4643. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GQR: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4643–4643. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GQE: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4643–4643. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 075S: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4643–4643. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack with affine diagonal. Let B be a ring. Let $F : QCoh(\mathcal{O}_\mathcal{X}) \to Mod_B$ be a right exact tensor functor which commutes with direct sums. Let $g : U \to \mathcal{X}$ be a morphism with $U = \mathrm{Spec}(A)$ affine. Then*

1. *$C = F(g_{QCoh,*}\mathcal{O}_U)$ is a commutative B-algebra and*
2. *there is a ring map $A \to C$*

such that $F \circ g_{QCoh,*} : Mod_A \to Mod_B$ sends M to $M \otimes_A C$ seen as B-module.

Proof. We note that g is quasi-compact and quasi-separated, see Morphisms of Stacks, Lemma 075S. In Cohomology of Stacks, Proposition 077A we have constructed the functor $g_{QCoh,*} : QCoh(\mathcal{O}_U) \to QCoh(\mathcal{O}_\mathcal{X})$. By Cohomology of Stacks, Remarks 0GQR and 0GQM we obtain a multiplication

$$\mu : g_{QCoh,*}\mathcal{O}_U \otimes_{\mathcal{O}_\mathcal{X}} g_{QCoh,*}\mathcal{O}_U \longrightarrow g_{QCoh,*}\mathcal{O}_U$$

which turns $g_{QCoh,*}\mathcal{O}_U$ into a commutative $\mathcal{O}_\mathcal{X}$-algebra. Hence $C = F(g_{QCoh,*}\mathcal{O}_U)$ is a commutative algebra object in Mod_B, in other words, C is a commutative B-algebra. Observe that we have a map $\kappa : A \to \mathrm{End}_{\mathcal{O}_\mathcal{X}}(g_{QCoh,*}\mathcal{O}_U)$ such that for any $a \in A$ the diagram

$$\begin{array}{ccc} g_{QCoh,*}\mathcal{O}_U \otimes_{\mathcal{O}_\mathcal{X}} g_{QCoh,*}\mathcal{O}_U & \xrightarrow{\mu} & g_{QCoh,*}\mathcal{O}_U \\ \downarrow{\scriptstyle \kappa(r)\otimes 1} & & \downarrow{\scriptstyle \kappa(r)} \\ g_{QCoh,*}\mathcal{O}_U \otimes_{\mathcal{O}_\mathcal{X}} g_{QCoh,*}\mathcal{O}_U & \xrightarrow{\mu} & g_{QCoh,*}\mathcal{O}_U \end{array}$$

commutes. It follows that we get a map $\kappa' = F(\kappa) : A \to \mathrm{End}_B(C)$ such that $\kappa'(a)(c)c' = \kappa'(a)(cc')$ and of course this means that $a \mapsto \kappa'(a)(1)$ is a ring map $A \to C$.

The morphism $g : U \to \mathcal{X}$ is affine, see Morphisms of Stacks, Lemma 0GQE. Hence $g_{QCoh,*}$ is exact and commutes with direct sums by Cohomology of Stacks, Lemma 0GQY. Thus $F \circ g_{QCoh,*} : \mathrm{Mod}_A \to \mathrm{Mod}_B$ is a right exact functor which commutes with direct sums and which sends A to C. By Functors and Morphisms, Lemma 0GNS we see that the functor $F \circ g_{QCoh,*}$ sends an A-module M to $M \otimes_A C$ viewed as a B-module. □